

UniConnect: A Student Services Portal for Kigali Tech University

Kigali Tech University (KTU) is a mid-sized university with approximately 8,000 students across undergraduate and postgraduate programmes. The university currently has no unified digital portal; students use a combination of email, a legacy notice board system, a separate library website, and in-person visits to the registrar's office for everything from course registration to fee payment.

The university's IT department has been commissioned to build UniConnect, a new all-in-one student services portal. The portal must serve students, academic staff (lecturers), and the registry (administrative staff). A mobile app is expected but the web platform is the primary deliverable for the first release.

A requirements analyst has conducted initial interviews with four stakeholders. Their transcripts are provided below. Your task is to analyse these interviews and extract, structure, and prioritise the requirements for the first release of UniConnect.

Important: The stakeholders were interviewed separately. They have not seen each other's responses. Some of them want contradictory things. Some of them use the same words to mean different things. Your job is to notice this.

Stakeholder Profiles

1. **Amina Nkosi 2nd year undergraduate**, Computer Science. Heavy mobile user. Frustrated by the current system. Represents the majority of student experience.
2. **Dr. Emmanuel Bizimana Senior Lecturer**, Engineering Faculty. Has been at KTU for 11 years. Cautious about technology. Represents academic staff.
3. **Grace Uwimana Head of Registry**. Manages course registrations, fee records, and official communications for all 8,000 students. Represents administrative staff.
4. **Patrick Mugabo IT Director**, KTU. Commissioning the system. Has a fixed budget and an 8-month delivery timeline. Represents technical and business constraints.

Stakeholder Transcripts: Amina Nkosi

"Honestly the biggest problem is just not knowing what's going on. Like, I'll find out there's a timetable change two days after it happened because someone posted it on the department WhatsApp group, not through any official channel. That's not okay. I need to know about things that affect my schedule in real time, not through gossip."

"Registration is a nightmare. Every semester it's the same thing you go to the registrar's office, you queue, you fill in a paper form, they lose it half the time, and then you find out three weeks in that you're not actually enrolled in a course you thought you were, I've had to redo a whole semester's worth of admin because of that. It should be completely online. Click, confirm, done"

finance office to get a receipt, and they manually update some spreadsheet somewhere. Why is the "I pay my fees through mobile money. It works fine for me. But then I have to go physically to the receipt not just emailed to me automatically? And why can I not see my fee balance and payment history in one place?"

live off campus so I use the portal mostly on my phone. Whatever you build, it has to work perfectly on mobile. I'm not going to carry a laptop everywhere just to register for a course. If it doesn't work on my phone it basically doesn't exist for me."

of their personal WhatsApp numbers and some I just have email, and half the time they don't "I also want to be able to message my lecturers directly through the system. Right now I have to respond. It would be great to have one place to reach everyone."

"Oh-and results. I want to see my results as soon as they're released, not find out from a friend who happened to check the notice board. And I want to see my full transcript, not just this semester's grades. I want to track my GPA over time."

Stakeholder Transcripts: Dr. Emmanuel Bizimana

"I want to be very clear about something from the beginning. I have been at this university for eleven years and I have seen many 'new systems' come and go. They are always introduced quickly, nobody is trained properly, and then the lecturers are expected to figure it out themselves while still teaching full courses. If this system is going to work, the rollout has to be phased and there has to be proper training. Not a PDF manual actual training."

"What I need is very simple. I need to be able to post my course materials slides, readings, assignments and have students access them in one place. At the moment I use email, I use a USB drive sometimes, I post on the department website which nobody maintains anymore. It's chaos. One place for everything."

"The thing I absolutely do not want is for students to be able to contact me at any time through the system. I already have too many emails. If there is a messaging function, I need to be able to set hours-like, messages only received between 8am and 5pm on weekdays and I need to be able to turn it off during exam marking. Students knowing they can reach me instantly will make things worse, not better."

"Results. This is sensitive. I do not want students to see their results the moment I submit them. There needs to be a moderation period sometimes results are adjusted after submission. especially for borderline cases or when there's been an error. If a student sees a grade that later changes, it causes enormous problems. Grades should only be visible after the faculty has signed them off. That process takes at least two weeks."

I also want to be able to see which students have registered for my courses each semester, and I want that list to be accurate and final before teaching starts. I have had students show up to my class who were never officially registered. That is an administrative problem but it creates problems for me."

"One more thing. I don't use my phone for work. Please don't make this system require a smartphone. I do everything on my office computer and I need the desktop experience to be fully functional, not an afterthought."

Stakeholder Transcripts: Grace Uwimana

"My team processes registration for 8,000 students every semester. At the moment we do most of it manually. Students come in, fill in forms, we enter the data into our spreadsheets, we check against the fee payment records, we check prerequisites, we issue confirmation. It takes my team of six people the entire first three weeks of semester to process. We can't do anything else during that time. If the system can automate even 70% of that, it changes everything for us."

"The most important thing is data integrity. When a student registers for a course, the system must check automatically: has this student paid their fees? Do they have the prerequisite courses? Is there a space in the course? All three checks must happen before registration is confirmed. At the moment, students sometimes slip through without paying we only catch it weeks later when we do our manual reconciliation. That creates a fee debt problem that is very difficult to recover."

that is non-"I need to be very careful about results visibility. Students should not be able to see results until the faculty has formally approved them and the registry has received the final, signed gradebook. In my experience, when students see provisional results they start contesting them informally, they go directly to lecturers, it creates chaos. Results visible only after official release negotiable for me."

about registration deadlines, fee "Communications. Right now I send official communications payment notices, academic calendar changes by printing notices and putting them on boards. Half the students never see them. I need to be able to send official announcements through the portal and know that students have actually received them. Ideally with a read-receipt or at least a delivery confirmation."

"One concern I have about online registration is errors. If we move to a self-service model, students will make mistakes wrong course, wrong semester, typos. I need a correction window-maybe 48 hours after registration - where students can change their selections, but after that window it should be locked and changes require my office to approve them. Otherwise it becomes a management problem."

"Fee payments we accept mobile money and bank transfers currently. The system should record all payments automatically, not require manual entry. And it should generate an official receipt immediately when payment is confirmed. I do not want students coming to our office for receipts anymore. That alone wastes two full days of my team's time every month."

Stakeholder Transcripts: Patrick Mugabo

"I'll be direct with you. We have a budget of \$120,000 USD for this project and a hard deadline of eight months. The university's academic calendar means we must go live before the start of the next registration period, which is fixed. If we miss that window, we wait another full year. There is no flexibility on either the budget or the timeline."

"The scope has to be realistic. I've seen the wishlist that's been floating around and some of it is simply not possible in this timeframe. A mobile app, a web portal, integration with our existing that is two or three years finance system, a messaging platform, a learning management system of work, not eight months. We need to decide what the first release actually is and stick to it."

"My priority for release one is: online course registration, fee payment visibility, and official announcements. Everything else course materials, messaging, the mobile app -two or later. I've been doing this for fifteen years. Scope creep kills projects." that is release

"The system has to integrate with our existing student information system, which runs on Oracle. That integration is complex and has cost implications. Any feature that requires reading from or writing to the Oracle database needs to go through a proper data architecture review. I do not want us building something that creates a data consistency problem."

"Security and privacy are non-negotiable. Student records are confidential. Results, fee balances, personal details - none of this should be accessible to anyone other than the student themselves and authorised staff. We need proper role-based access control from day one. This is not optional."

"One last thing. The university has a lot of staff who are not technically sophisticated. Whatever we build needs to work without training for basic functions. If Grace's team needs a three-day course to send an announcement, we've built the wrong thing. Self-explanatory interfaces, proper error messages, no jargon."

YOUR TASKS:

1. Read all four stakeholder transcripts carefully. As a group, extract at least 6 user stories from the material. Each story must:

Use the format: As a [specific role], I want [goal], so that [benefit].

Name a specific role from the transcripts (not just "user" or "admin")

State a real benefit in the "so that" clause not just a restatement of the goal

Pass the INVEST criteria check each story against all six criteria and revise any that fail

Aim for stories from at least three different stakeholder perspectives. You may write more than 6.

2. Write 2 acceptance criteria in Given/When/Then format for each user story identified in the material.